10/28/2025

Programming Fundamentals

Assignment #2

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Section: BCS-1k
Assignment #2
Course: Programming Fundamentals
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Note: This PDF contains only the screenshots of the outputs of the code and programs themselves, not the code. The codes are uploaded to GCR in a .ZIP file, as well as my GitHub account link where I post all my PF Assignment Codes and PF Lab Tasks.

Therefore, for the code itself, please download and open the .ZIP file or go to my GitHub account, both of which are uploaded to GCR alongside this PDF.

The Directions for the screenshots if there is more than one on the slide at once is: Read from Left to Right.

Question 1:

```
1. Add a New Book.
2. Process a Sale.
3. Generate Low-Stock Report.
4. Exit the Program.
Enter your Choice.
1

Enter the book's ISBN.
1234
Enter the Book's Title:
C Learning
Enter the Book's Price.
9.99
Enter the quantities of the book.
10
The book has successfully been added.

1. Add a New Book.
2. Process a Sale.
3. Generate Low-Stock Report.
4. Exit the Program.
Enter your Choice.
1

Enter the book's ISBN.
2345
Enter the Book's Title:
Introduction to Linux
Enter the Book's Price.
12.99
Enter the quantities of the book.
30
The book has successfully been added.

1. Add a New Book.
2. Process a Sale.
3. Generate Low-Stock Report.
4. Exit the Program.
Enter your Choice.
2
```

```
Enter your Choice.
2

Enter the ISBN of the Book.
1234
Enter the number of copies to be sold.
8
Sale successful.

1. Add a New Book.
2. Process a Sale.
3. Generate Low-Stock Report.
4. Exit the Program.
Enter your Choice.
2

Enter the ISBN of the Book.
1234
Enter the number of copies to be sold.
5
Insufficient Stock, please try again.
The remaining stock of this book is 2.


Enter the ISBN of the Book.
2345
Enter the number of copies to be sold.
10
Sale successful.

1. Add a New Book.
2. Process a Sale.
3. Generate Low-Stock Report.
4. Exit the Program.
Enter your Choice.
3
```

```
1. Add a New Book.
2. Process a Sale.
3. Generate Low-Stock Report.
4. Exit the Program.
Enter your Choice.
3

 ISBN:        1234
 Title:       C Learning
 Price:         9.99
 Quantities:  2

1. Add a New Book.
2. Process a Sale.
3. Generate Low-Stock Report.
4. Exit the Program.
Enter your Choice.
4
PS F:\CodeFromVS\Code\PFTheory\PFTheoryAssignment2>
```

Question 2:

```
1. Enter Customer Information.
2. Display Inventory.
3. Add Items to Cart.
4. Display Total Bill and Invoice.
5. Exit the Program.
Enter your choice.
1

Enter Customer Name.
Muhammad Farhan

Enter Customer CNIC.
0538


1. Enter Customer Information.
2. Display Inventory.
3. Add Items to Cart.
4. Display Total Bill and Invoice.
5. Exit the Program.
Enter your choice.
2

Product Code    Quantity in Stock       Price per Product.

1               50                      100
2               10                      200
3               20                      300
4               8                       150

1. Enter Customer Information.
2. Display Inventory.
3. Add Items to Cart.
4. Display Total Bill and Invoice.
5. Exit the Program.
Enter your choice.
3
```

```
Enter your choice.
3


Product ID      Price
1               100
2               200
3               300
4               150
Enter product ID to add to cart.
1
Enter quantity.
10

1. Add another Product.
2.Go to Main Menu.
Enter your choice.
1

Product ID      Price
1               100
2               200
3               300
4               150
Enter product ID to add to cart.
3
Enter quantity.
20

1. Add another Product.
2.Go to Main Menu.
Enter your choice.
1

Product ID      Price
1               100
2               200
3               300
4               150
Enter product ID to add to cart.
4
```

```
Enter product ID to add to cart.
4
Enter quantity.
15
Not enough stock. Only 8 available.

1. Add another Product.
2.Go to Main Menu.
Enter your choice.
2


1. Enter Customer Information.
2. Display Inventory.
3. Add Items to Cart.
4. Display Total Bill and Invoice.
5. Exit the Program.
Enter your choice.
4


Customer Name: Muhammad Farhan

Customer CNIC: 538
Product Code    Product Quantity        Price per Item
1               10                      100
3               20                      300
Total Bill:     7000.00

1. Enter promocode.
2. Complete Order.
Enter your Choice.
1

Enter your promocode
Eid202
Invalid Promocode, Please try again.
```

```
Enter your promocode
Eid202
Invalid Promocode, Please try again.


1. Enter promocode.
2. Complete Order.
Enter your Choice.
1

Enter your promocode
Eid2025

Promocode Accepted, 25 percent discount applied.


        The Final Invoice

Customer Name: Muhammad Farhan

Customer CNIC: 538
Product Code    Product Quantity        Price per Item
1               10              100
3               20              300
Discount after Promocode: 1750.00
Total Bill:     5250.00

1. Enter Customer Information.
2. Display Inventory.
3. Add Items to Cart.
4. Display Total Bill and Invoice.
5. Exit the Program.
Enter your choice.
5

PS F:\CodeFromVS\Code\PFTheory\PFTheoryAssignment2>
```

Question 3:

```
1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
3

Total number of Overloaded Sectors: 11
Total number of Sectors that require maintainance: 10


1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
2

Specify the coordinates of your sector of the 5x5 Grid in the exact format: (row,column) (ex: (2,3)).
(1,2)

Power Status: OFF
Overload Warning: Overloaded
Maintainance Required: Yes


1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
1

Specify the coordinates of your sector of the 5x5 Grid in the exact format: (row,column) (ex: (2,3)).
(1,2)

1. Set a specific status.
2. Clear a specific status.
Enter your choice.
```

```
1. Set a specific status.
2. Clear a specific status.
Enter your choice.
1

1. Power Status.
2. OverLoad Warning.
3. Maintainance Required.
Enter your choice.
1

The selected status on sector (1, 2) has been Set.



1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
1

Specify the coordinates of your sector of the 5x5 Grid in the exact format: (row,column) (ex: (2,3)).
(1,2)

1. Set a specific status.
2. Clear a specific status.
Enter your choice.
2

1. Power Status.
2. OverLoad Warning.
3. Maintainance Required.
Enter your choice.
3

The selected status on sector (1, 2) has been Cleared.



1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
```

```
The selected status on sector (1, 2) has been Cleared.



1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
2

Specify the coordinates of your sector of the 5x5 Grid in the exact format: (row,column) (ex: (2,3)).
(1,2)

Power Status: ON
Overload Warning: Overloaded
Maintainance Required: No


1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
3

Total number of Overloaded Sectors: 11
Total number of Sectors that require maintainance: 9


1. Update Sector Status.
2. Query Sector Status.
3. Run System Diagnostic.
4. Exit the Program.
Enter your choice.
4
PS F:\CodeFromVS\Code\PFTheory\PFTheoryAssignment2>
```

Question 4:

```
Enter the size of your Stack.
5

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
1

Enter the integer to Push
5
The element has been pushed.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
1

Enter the integer to Push
7
The element has been pushed.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
1

Enter the integer to Push
3
The element has been pushed.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
```

```
The element has been pushed.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
4

0
0
3
7
5

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
3
The top stack is 3.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
2
The element has been popped.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
4
```

```
Enter your choice.
4

0
0
0
7
5

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
1

Enter the integer to Push
1
The element has been pushed.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
1

Enter the integer to Push
2
The element has been pushed.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
1

Enter the integer to Push
3
The element has been pushed.
```

```
The element has been pushed.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
1

Enter the integer to Push
4
The element cannot be pushed, the stack is full.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
2
The element has been popped.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
2
The element has been popped.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
2
The element has been popped.
```

```
The element has been popped.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
2
The element has been popped.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
2
The element has been popped.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
2
The stack is empty, it cannot be popped.

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
4

0
0
0
0
0
```

```
Enter your choice.
4

0
0
0
0
0

1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice.
5
PS F:\CodeFromVS\Code\PFTheory\PFTheoryAssignment2>
```

Question 5:

```
1. Encode Message.
2. Decode Message.
3. Exit.
Enter your choice.
1

Enter your Message.
Farhan BCS-1K 25K-0538

The Encoded message is:
*!'"?Y' 2Y#?AQP2|sz`sT


1. Encode Message.
2. Decode Message.
3. Exit.
Enter your choice.
2

Enter your Encoded Message.
*!'"?Y' 2Y#?AQP2|sz`sT

The Decoded message is:
Farhan BCS-1K 25K-0538


1. Encode Message.
2. Decode Message.
3. Exit.
Enter your choice.
1

Enter your Message.
The Rider is Ready!

The Encoded message is:
3kvsw@2a{2`wv{@2wzF
```

```
The Encoded message is:
3kvsw@2a{2`wv{@2wzF


1. Encode Message.
2. Decode Message.
3. Exit.
Enter your choice.
2

Enter your Encoded Message.
3kvsw@2a{2`wv{@2wzF

The Decoded message is:
The Rider is Ready!


1. Encode Message.
2. Decode Message.
3. Exit.
Enter your choice.
3
```